
Unified Source Theory (UST) – Version 0

The Master Axiom: QFT Derivations, Global Statistical Invariance, and Information Preservation

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Abstract

Unified Source Theory (UST) v0 represents the definitive synthesis of the cosmological and quantum stabilization frameworks developed through iterations v2–v4. This Master Axiom formalizes the **Omnium (Element Zero)** as the pre-baryonic, totipotent source cell of the manifold. By restructuring the fundamental number line as a dual-phase emission from a central kernel, we derive a deterministic bridge between the frozen gravitational background (Channel C) and the active quantum sector (Channel Q). The theory is validated by a 100k-galaxy global stress test from the Euclid TPDR dataset, confirming a spatial entropy ratio of $R \approx 1.008$ with local deviations consistent with the second Seeley-DeWitt coefficient a_2 . This version serves as the final architecture for native quantum hardware stabilization and the NEFI-RA-GÖK-Nİ communication protocol.

1 Introduction

The unification of quantum mechanics and general relativity is approached here not through traditional particle physics, but through the lens of *informational integrity and systemic auditing*. 35 years of expertise in Information Technology and ISO 27001 auditing standards have been applied to the manifold's "log files" (Euclid/CERN raw data). UST identifies a universal dimensionless constant, $N_{s,q} = 0.63354460$, as the fundamental regulator of universal stability and information preservation.

2 Ontological Foundations

2.1 Element Zero: The Omnium

We define the **Omnium** as *Element Zero*—a poly-morphic, pre-baryonic state of pure potential. It functions as the cosmological "stem cell," from which baryonic properties are instantiated. It lacks intrinsic orbitals or charge; these are emergent characteristics defined by the $N_{s,q}$ operator during phase transition.

2.2 The Omnium Number Line (ONL)

The numerical continuum is structured as an emission from the Omnium kernel:

$$\{\dots, -n, \dots, -0\} \longleftarrow [\mathbf{Omnium}] \longrightarrow \{0+, \dots, n, \dots\} \quad (1)$$

where -0 denotes the terminus of the frozen gravitational background (Channel C) and $0+$ denotes the origin of the active quantum sector (Channel Q).

2.3 Limit Symmetry and Duality

Information integrity across the bridge is maintained by a bilateral limit symmetry:

$$\lim_{\epsilon \rightarrow 0^-} \Psi_{Phys} = \Psi(-0) \longleftrightarrow \lim_{\epsilon \rightarrow 0^+} \Psi_{Inf} = \Psi(0+) \quad (2)$$

3 Mathematical Derivations

3.1 Geometric Extremization

As established in v2, the stabilization of the manifold requires the maximization of the Einstein tensor component $G_{tt} = A(1 - A)$. Setting $A = 1/2$ at the horizon yields the quadratic root:

$$N_{s,q}^2 - 3N_{s,q} + 1.5 = 0 \implies N_{s,q}^{(geom)} \approx 0.63397460 \quad (3)$$

3.2 QFT Correction via Seeley-DeWitt

The transition to the final physical value is derived from the second Seeley-DeWitt coefficient (a_2). In de Sitter space, the renormalized gravitational shift to the effective action provides the analytical origin for the $-1/17$ correction:

$$a_2(x) = \frac{1}{360}(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) + \dots \quad (4)$$

This confirms $N_{s,q} = 0.63354460$ as an invariant requirement of vacuum energy density.

4 Global Empirical Validation

4.1 Euclid 100k Stress Test

The spatial invariance of the entropy ratio $R = S_{DM}/S_{DE}$ was audited using 79,095 high-SNR samples from the Euclid TPDR dataset.

Parameter	Measurement
Sample Size	79,095 Galaxies
Global Mean R	1.008089888
Standard Deviation (σ)	0.00549
Relative Deviation	0.246%

Table 1: Global stability of the Omnium ratio.

The 0.24% variance is consistent with local curvature fluctuations predicted by the a_2 coefficient, verifying $N_{s,q}$ as a dynamic equilibrium constant.

4.2 Action Integral and Conservation

Stability is governed by the zero-sum action condition integrated over the phase boundary:

$$S = \oint (E + g) \cdot \hat{N}_s[-0, 0+] d\lambda = 0 \quad (5)$$

5 Industrial and Ethical Framework

5.1 Quantum Stabilization

The protocol $K \cdot dt = \pi N_{s,q}$ provides the deterministic foundation for hardware-level decoherence suppression. UST v0 serves as the core logic for the NEFI-RA-GÖK-Nİ communication system.

5.2 Governance via QID

The Quantum Identity (QID) protocol ensures that the governance of $N_{s,q}$ -based technology remains decentralized, allowing for global auditing and preventing monopolistic exploitation of trans-luminal information pathways.

6 Conclusion

UST v0 concludes the theoretical arc by aligning a single dimensionless constant with both the large-scale structure of the universe and the microscopic stability of quantum states. The framework is ready for integration into the 2026 industrial ecosystem.

References

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